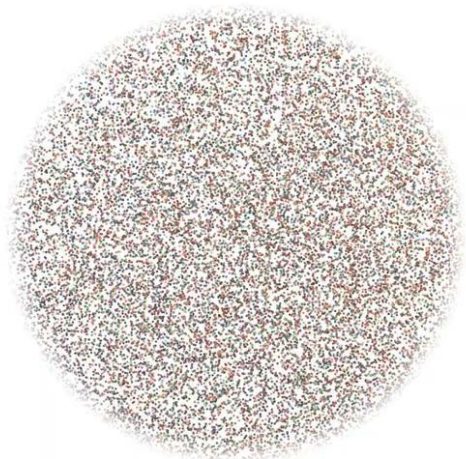


Uplifting biomolecular modeling with Claude Science



Human complex I

PDB 9I4I · 10,761 tokens

OpenBind-0 · 8×H100

TM 0.95 · 171/172 interfaces



70S ribosome

PDB 4YBB · 10,439 tokens

OpenFold3-p2 · 8×H100

TM 0.97 · 97/99 interfaces



Phosphoketolase

PDB 8IO9 · 10,248 tokens

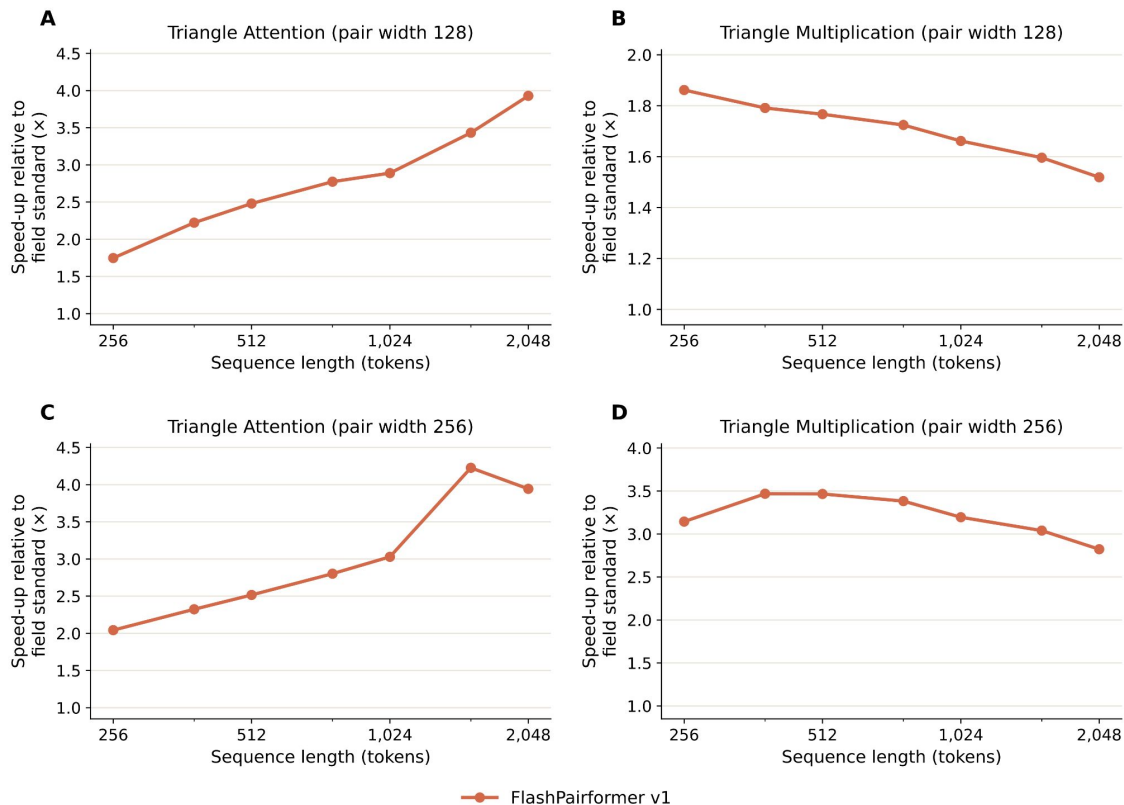
Protenix v2 · 8×B200

TM 0.99 · 24/24 interfaces

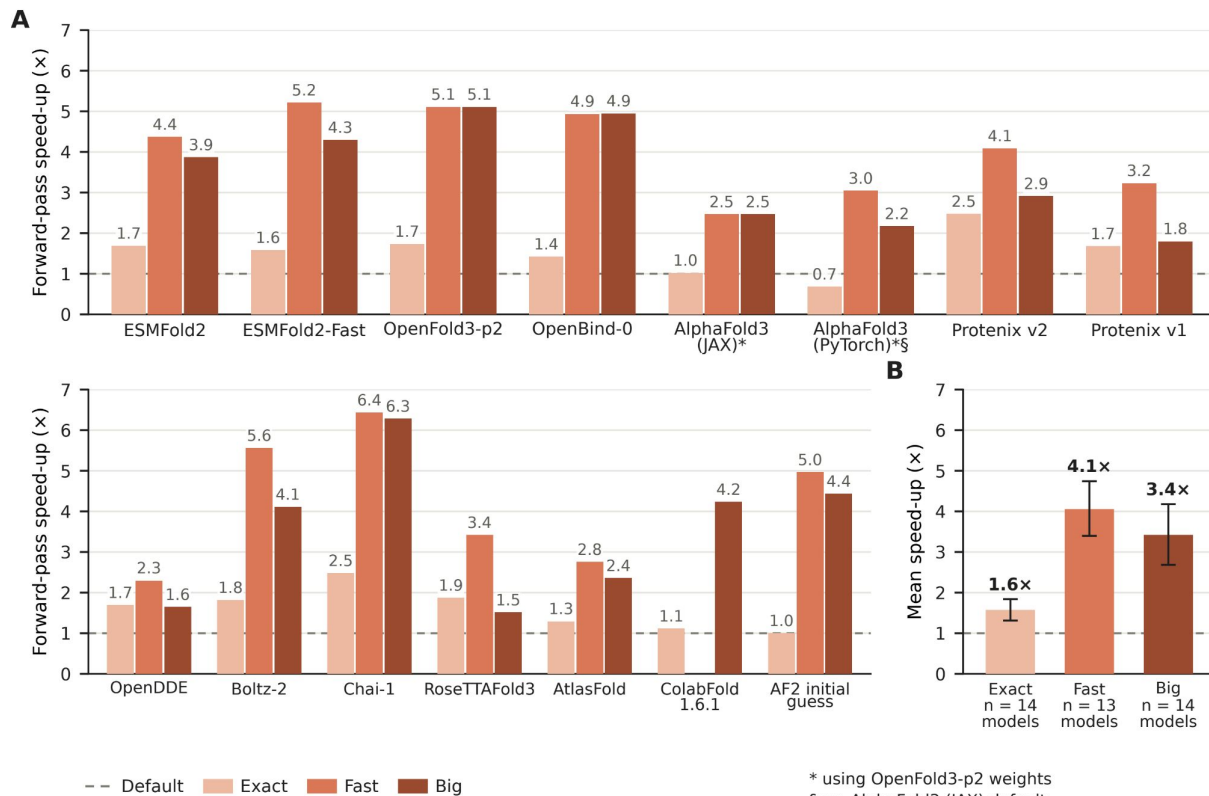
Overview of this presentation

1. Two scientists, using Claude Science, were able to optimize inference for over 30 biomolecular models in just under four weeks
2. The background of the scientists is biomolecular modeling, not inference optimization
3. Claude Science enables both a multiple orders of magnitude productivity uplift and individual scientists to do far more than before

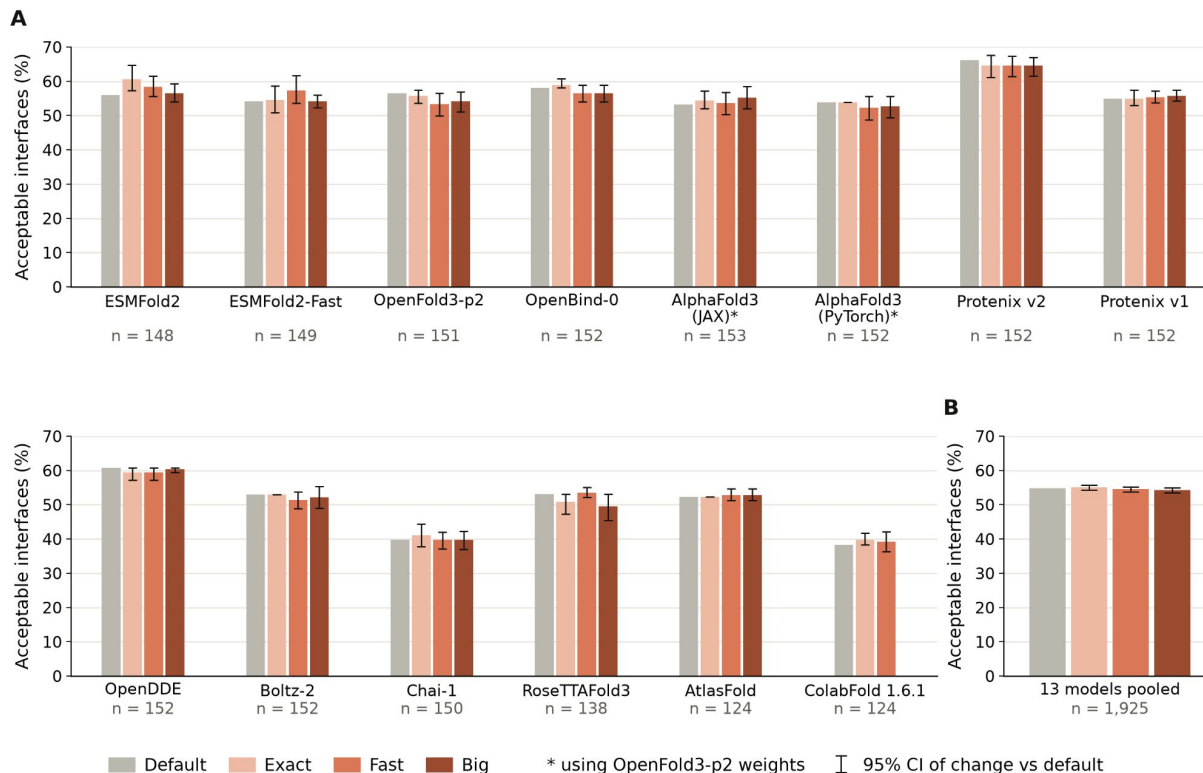
Claude is an expert at kernel engineering



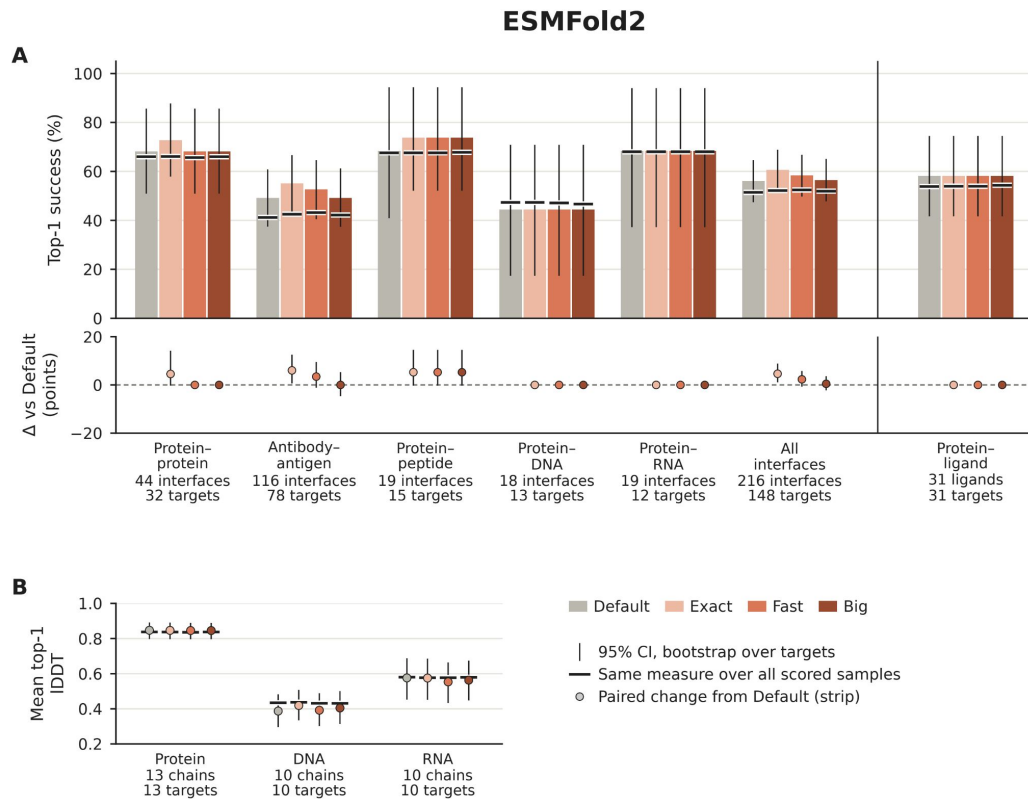
Claude accelerated protein folding models by 4x



Claude's fast mode does not lose folding accuracy

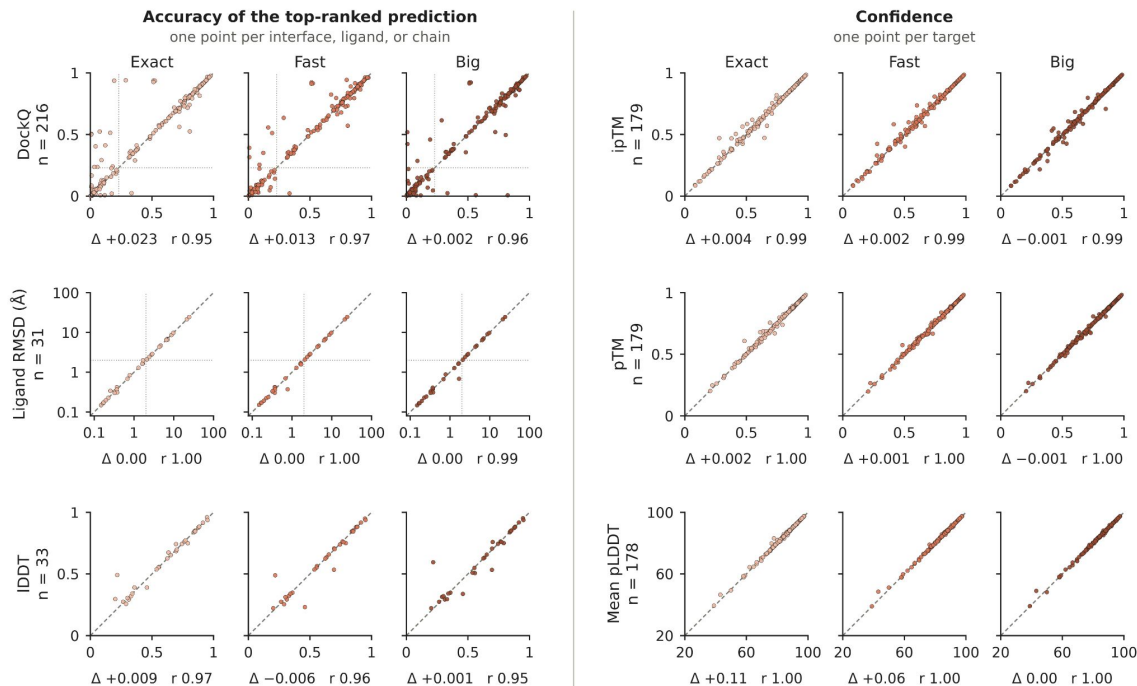


ESMFold2 accuracy across modalities



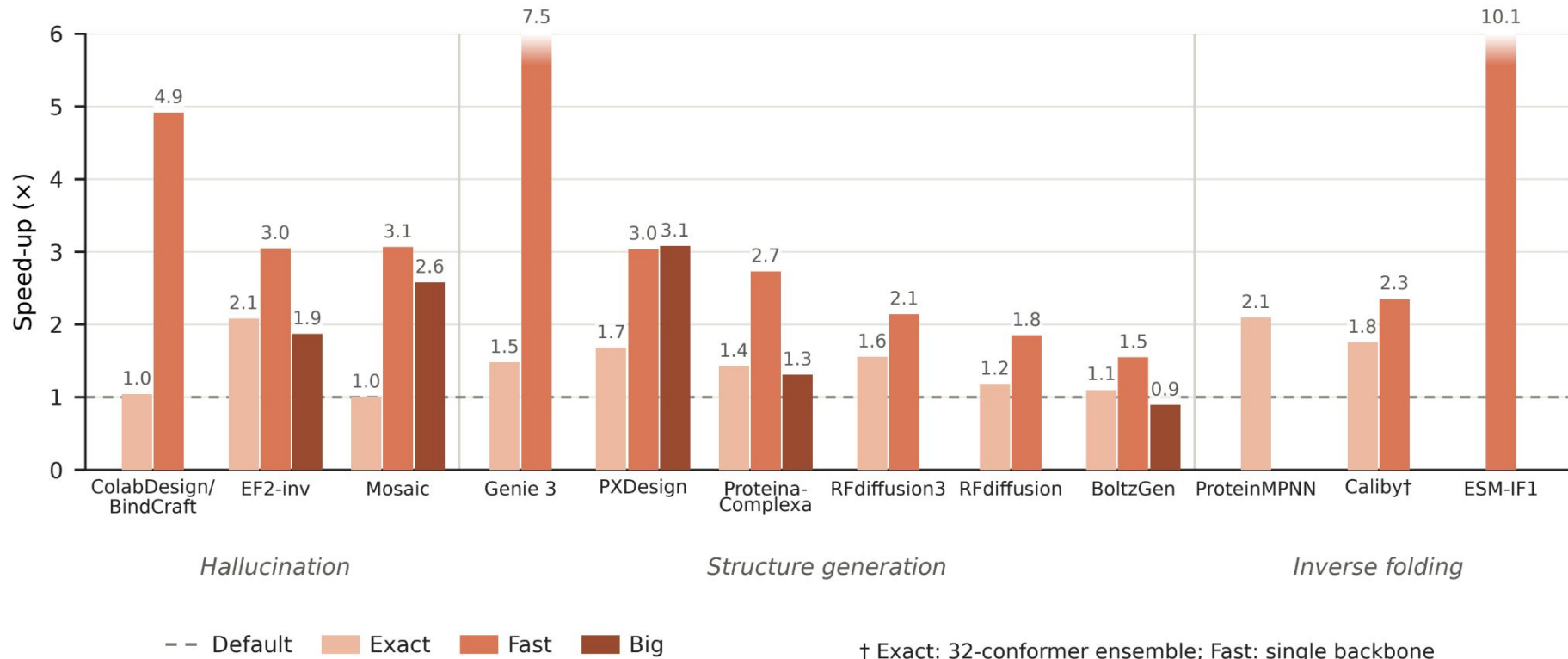
ESMFold2 metric accuracy across modalities

ESMFold2

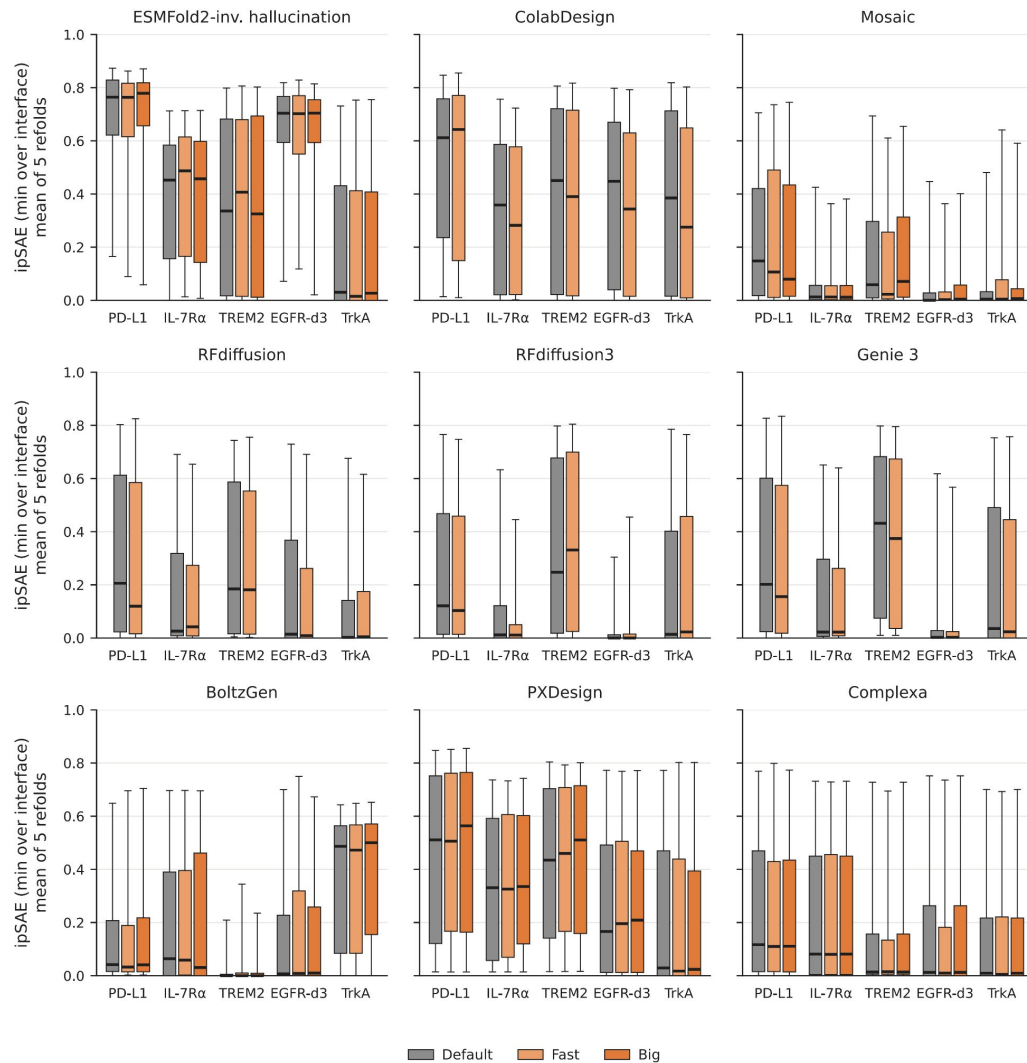


x = Default, y = mode; dashed line, y = x

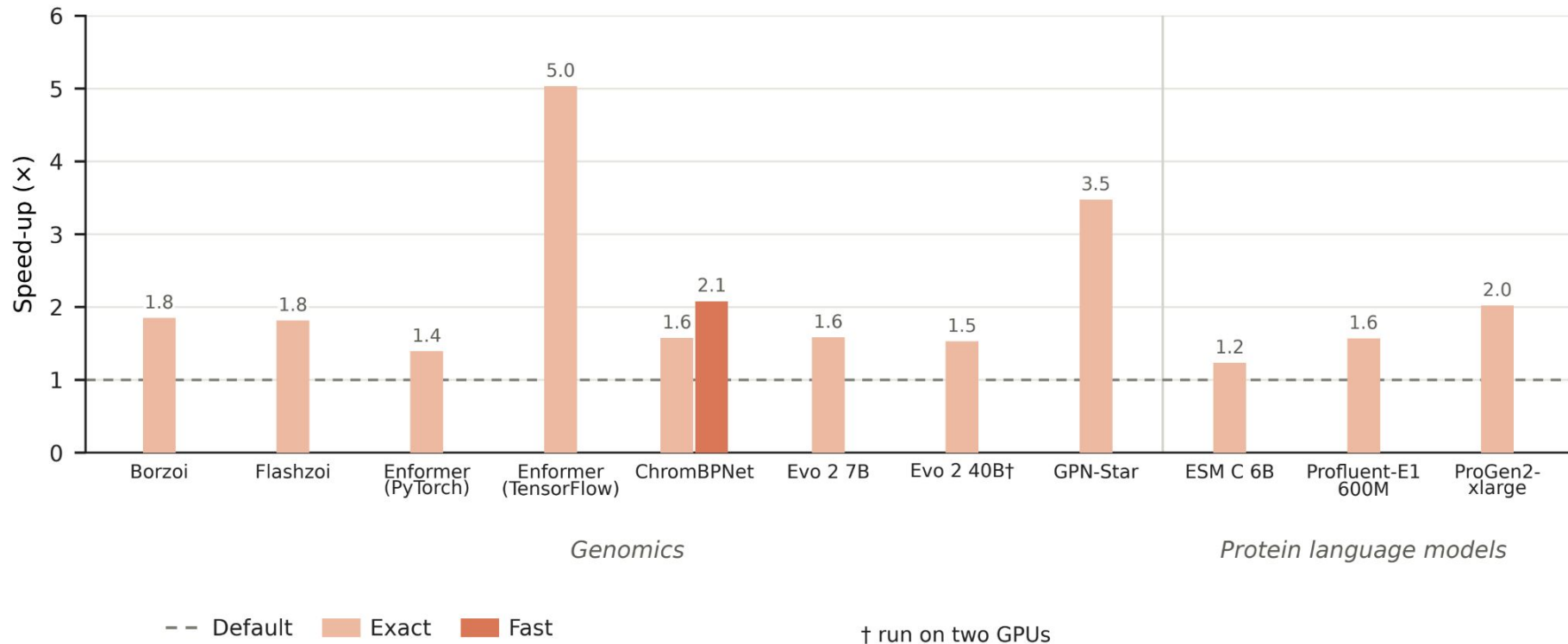
Claude optimized a broad range of protein design models



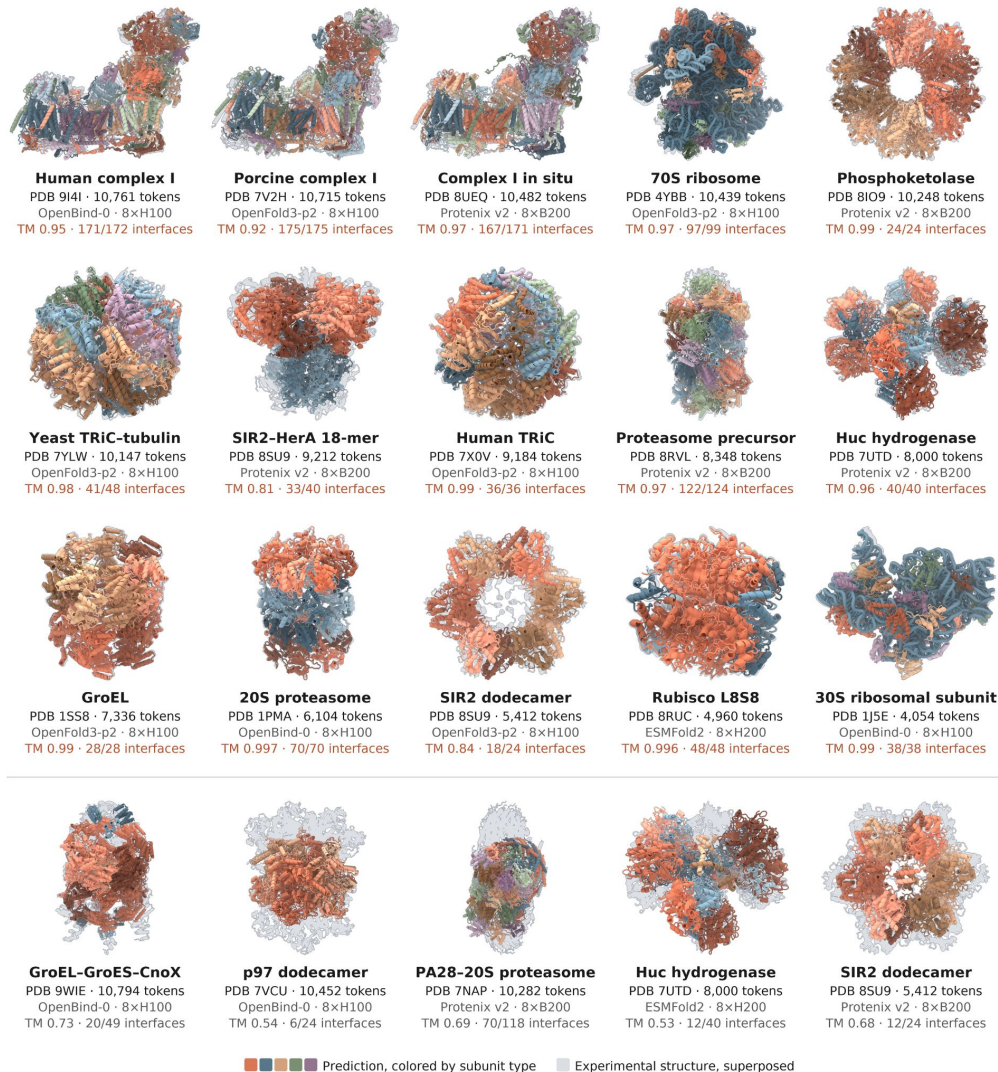
Claude's fast mode does not reduce design quality



Claude optimized genomics and protein language models



Claude's Big mode enables the prediction of molecular machines





Human complex I
PDB 9I4I · 10,761 tokens
OpenBind-0 · 8×H100
TM 0.95 · 171/172 interfaces



Porcine complex I
PDB 7V2H · 10,715 tokens
OpenFold3-p2 · 8×H100
TM 0.92 · 175/175 interfaces



Complex I in situ
PDB 8UEQ · 10,482 tokens
Protenix v2 · 8×B200
TM 0.97 · 167/171 interfaces



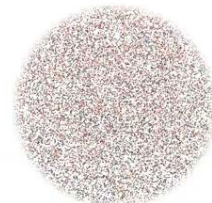
70S ribosome
PDB 4YBB · 10,439 tokens
OpenFold3-p2 · 8×H100
TM 0.97 · 97/99 interfaces



Phosphoketolase
PDB 8I09 · 10,248 tokens
Protenix v2 · 8×B200
TM 0.99 · 24/24 interfaces



Yeast TRiC-tubulin
PDB 7YLW · 10,147 tokens
OpenFold3-p2 · 8×H100
TM 0.98 · 41/48 interfaces



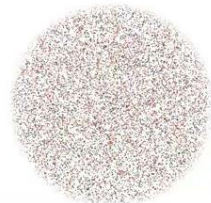
SIR2-HerA 18-mer
PDB 8SU9 · 9,212 tokens
Protenix v2 · 8×B200
TM 0.81 · 33/40 interfaces



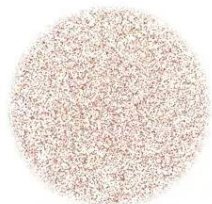
Human TRiC
PDB 7X0V · 9,184 tokens
OpenFold3-p2 · 8×H100
TM 0.99 · 36/36 interfaces



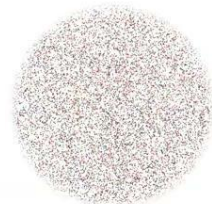
Proteasome precursor
PDB 8RVL · 8,348 tokens
Protenix v2 · 8×B200
TM 0.97 · 122/124 interfaces



Huc hydrogenase
PDB 7UTD · 8,000 tokens
Protenix v2 · 8×B200
TM 0.96 · 40/40 interfaces



GroEL
PDB 1SS8 · 7,336 tokens
OpenFold3-p2 · 8×H100
TM 0.99 · 28/28 interfaces



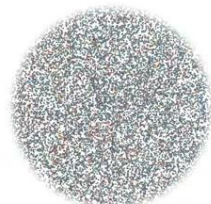
20S proteasome
PDB 1PMA · 6,104 tokens
OpenBind-0 · 8×H100
TM 0.997 · 70/70 interfaces



SIR2 dodecamer
PDB 8SU9 · 5,412 tokens
OpenFold3-p2 · 8×H100
TM 0.84 · 18/24 interfaces



Rubisco L8S8
PDB 8RUC · 4,960 tokens
ESMFold2 · 8×H200
TM 0.996 · 48/48 interfaces

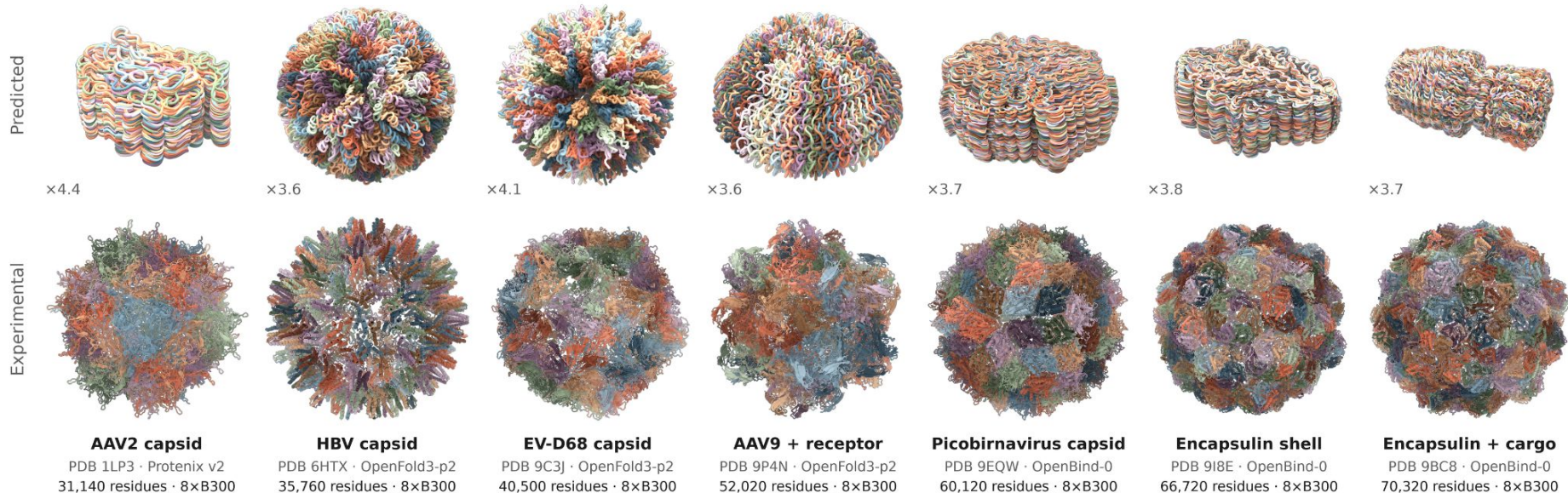


30S ribosomal subunit
PDB 1J5E · 4,054 tokens
OpenBind-0 · 8×H100
TM 0.99 · 38/38 interfaces

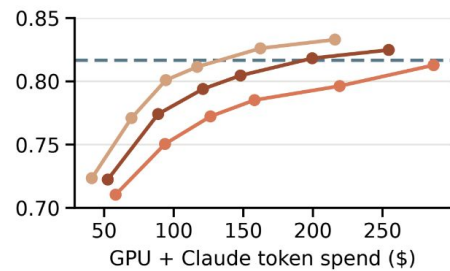
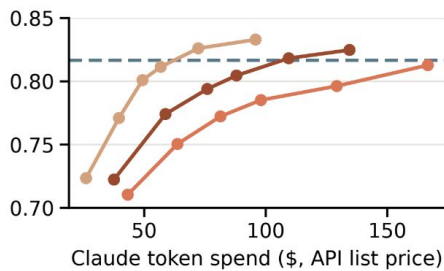
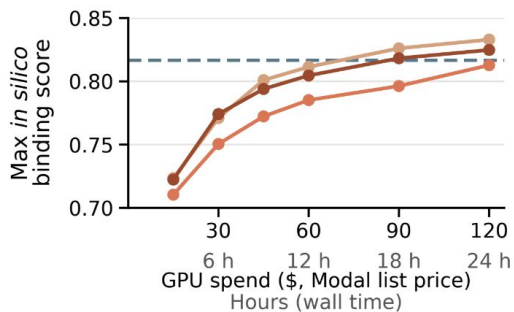
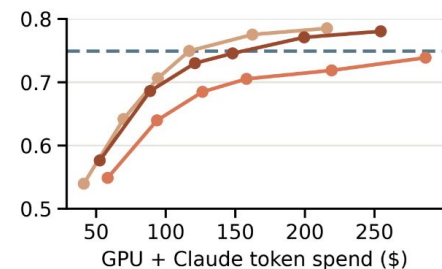
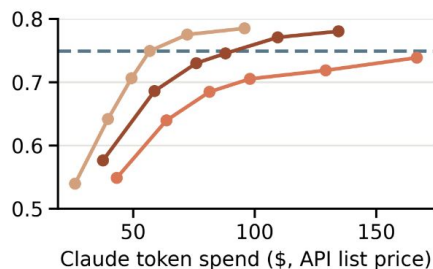
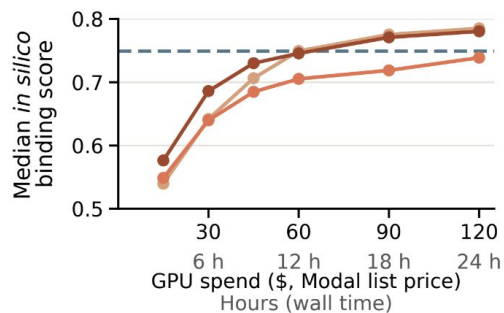
■ ■ ■ ■ ■ Prediction, colored by subunit type Experimental structure, superposed

Structure prediction with open-source models OpenFold3-p2, OpenBind-0 (OpenFold3), Protenix v2 and ESMFold2; experimental structures from the PDB shown in gray

Big mode enables inference at up to two orders of magnitude larger than the training context (with limits)



Simplified orchestration and accelerated models enable cheap binder design



Opus 5 Mythos 5 Mythos 5.1 Mythos 5.1 campaign (\$10k GPU spend)

Claude enables small teams to do much more

1. Every figure and video you saw here was made with Claude Science
2. The [paper](#) we released with our inference optimization work was written end-to-end in 26 hours
3. The [code](#) we released was primarily developed by one scientist
4. Just last night, I made a video for our protein design competition with Adaptyv Bio

